

Dhruv Jain

Portfolio | LinkedIn | GitHub | Twitter | dhruvrakeshjain@gmail.com | +91-7972205099

SUMMARY

Software Engineer Intern and Computer Science student specializing in Artificial Intelligence and Machine Learning, with hands-on experience developing full-stack B2B software, AI-driven applications, and real-time systems spanning backend infrastructure, data pipelines, and intelligent retrieval solutions.

EDUCATION

A.P. Shah Institute of Technology

B.E. CSE, Artificial Intelligence and Machine Learning

Thane, India

Aug 2023 – Present

EXPERIENCE

Software Engineer Intern

Jan 2026 – Present

Wallnut

Remote

- Architected and developed a full-stack B2B enterprise sales platform from scratch, engineering an automated data pipeline to parse, normalize, and sync multi-year Tally ERP ledger and voucher records into PostgreSQL.
- Implemented multi-tier Role-Based Access Control (RBAC) across 4 organizational levels (CEO, State Head, District Manager, Sales Officer), dynamically scoping financial KPIs, territory sales, and party outstandings.
- Engineered real-time sales analytics engines and invoice-level product drill-down modals, calculating Net Sales (excl. GST), YoY growth comparisons, and dealer performance metrics with sub-second response times.
- Digitized field sales operations by building an end-to-end sales call logging and daily reporting module, completely replacing manual spreadsheets in preparation for company-wide rollout across sales teams.

PROJECTS

MealSwitch | FastAPI, Python, React, Gemini API, Scikit-learn, PostgreSQL | Live

GitHub

- Engineered a content-based recommendation engine using Scikit-learn TF-IDF vectorization & Cosine Similarity over nutritional feature spaces to compute low-glycemic, macro-optimized food substitutions.
- Integrated Gemini LLM with defensive JSON schema validation, prompt guardrails, and asynchronous timeout/retry fallbacks to prevent token hallucinations and downstream parsing crashes.
- Architected RESTful backend with FastAPI & PostgreSQL (SQLAlchemy), integrating USDA FoodData APIs with JWT role-based access control and sliding-window rate-limiting.

Knowrex | Next.js, TypeScript, Pinecone, RAG, Redis, Supabase, Docker | Live

GitHub

- Architected a full-stack RAG support platform using Next.js and LLM orchestration, streaming source-cited responses with sub-2s end-to-end latency.
- Engineered a semantic retrieval pipeline with Pinecone vector DB using 384-dim dense embeddings and a self-learning loop for resolved edge-case queries.
- Designed confidence-based failure handling to route low-relevance queries to human support, backed by dual-tier Redis caching and 3-tier exponential backoff.

SalonWallah | React Native, TypeScript, Node.js, MongoDB, Redis, WebSockets, AWS EC2 | Live

GitHub

- Architected a dual-engine queue system using atomic state transitions, eliminating chair booking collisions and reducing customer wait times to 8 minutes.
- Implemented commute-aware dispatching via MongoDB geospatial queries (2dsphere), auto-alerting users 10 minutes prior to turn to minimize no-shows.
- Developed dual client and stylist apps in React Native (Expo) featuring Socket.io live syncing, Razorpay micro-payments, and AWS EC2 deployment.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL

Frontend: React, Next.js, React Native, Expo, Tailwind CSS

Backend & Data: FastAPI, Node.js, Express, MongoDB, PostgreSQL, Redis

Cloud & Developer Tools: AWS, Docker, Git, GitHub, Postman

ACHIEVEMENTS

TenzorX 2026 — Advanced to the Detailed Prototype Round (27K+ students)

April 2026